

ARAR, Saudi Arabia

WMO: 403570

Lat: 30.907N

Lon: 41.138E

Elev: 1813

StdP: 13.76

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.6	36.0	16.0	13.2	47.2	19.8	15.8	54.9	27.3	52.9	24.2	54.3	6.1	280	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	28.4	113.3	74.3	111.2	73.2	109.2	72.1	77.0	108.2	75.4	106.7	74.0	104.6	8.1	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	110.3	86.8	66.1	102.8	86.9	64.2	96.3	87.3	41.8	108.2	40.3	107.0	38.9	104.6	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.2	21.0	18.7	DB	28.5	116.6	3.7	2.1	25.9	118.2	23.8	119.4	21.7	120.6	19.0	122.1	
			WB	26.0	76.6	3.9	4.2	23.1	79.6	20.8	82.1	18.6	84.4	15.8	87.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	73.9	50.8	55.0	63.1	72.9	82.8	90.3	94.4	94.7	88.4	78.1	62.1	52.8
	DBStd	16.80	4.89	6.06	6.62	6.53	5.80	3.85	3.79	3.72	4.58	6.13	6.79	5.72
	HDD50	104	47	19	1	0	0	0	0	0	0	0	3	34
	HDD65	1364	441	284	116	10	0	0	0	0	0	2	133	378
	CDD50	8829	72	160	408	687	1018	1208	1377	1387	1152	872	365	123
	CDD65	4609	0	4	57	247	553	758	912	922	702	408	45	1
	CDH74	71104	7	81	746	2983	7819	12261	15664	15756	10586	4740	435	26
	CDH80	47028	0	13	249	1395	4717	8331	11299	11391	6966	2547	118	2
Wind	WSAvg	8.9	8.2	9.1	10.0	10.0	9.9	10.0	10.0	8.7	7.9	7.9	7.4	7.4
Precipitation	PrecAvg	1.9	0.4	0.3	0.2	0.3	0.1	0.0	0.0	0.0	0.0	0.2	0.3	0.3
	PrecMax	5.7	0.9	2.0	1.3	2.3	0.7	0.0	0.0	0.0	0.0	1.0	1.7	2.2
	PrecMin	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.1	0.2	0.4	0.3	0.5	0.2	0.0	0.0	0.0	0.0	0.2	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.4	81.0	93.5	99.0	106.4	113.0	116.3	116.5	111.6	102.5	88.2	77.3
		MCWB	57.4	59.0	64.4	65.8	69.9	73.2	76.1	76.2	73.6	70.5	64.8	57.4
	2%	DB	69.2	75.6	86.1	95.2	103.2	109.4	113.3	113.4	109.1	99.0	82.8	71.8
		MCWB	55.2	58.1	61.8	64.5	68.6	72.0	74.3	74.6	73.0	67.2	62.3	56.4
	5%	DB	64.7	71.8	82.0	91.6	100.5	107.2	111.2	111.4	105.9	96.6	78.9	67.8
		MCWB	53.7	56.1	60.2	64.0	67.4	70.3	72.9	73.5	71.1	66.2	61.4	56.3
	10%	DB	62.2	67.9	77.3	88.0	98.1	104.3	109.0	109.2	102.6	92.9	75.1	64.3
		MCWB	52.3	54.9	58.1	62.9	66.8	68.8	71.4	72.5	69.1	65.9	60.2	54.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.4	65.0	68.5	71.9	74.7	76.3	78.3	79.1	77.3	73.7	69.5	64.5
		MCDB	66.7	73.6	80.5	89.0	97.2	107.9	110.1	110.2	106.3	90.5	76.4	70.4
	2%	WB	57.5	61.0	65.0	69.0	72.0	74.1	76.3	77.4	74.8	71.4	66.6	60.4
		MCDB	64.7	70.9	80.9	85.8	96.6	105.1	108.2	109.6	103.5	91.2	76.8	65.9
	5%	WB	55.4	58.1	62.4	66.7	70.1	72.4	74.9	75.7	73.0	69.2	64.1	57.9
		MCDB	62.3	67.7	77.4	83.4	94.4	102.9	107.7	108.0	101.9	88.4	73.9	64.5
	10%	WB	53.4	55.8	59.8	64.6	68.4	70.8	73.4	73.9	71.3	67.5	62.1	55.7
		MCDB	61.0	65.8	74.5	83.6	92.2	101.1	105.8	105.8	99.8	87.1	71.1	62.6
Mean Daily Temperature Range	5% DB	MDBR	21.9	23.1	24.8	25.8	26.2	27.6	27.7	28.4	28.1	25.9	22.7	22.2
		MCDBR	25.7	27.8	29.2	29.2	28.4	30.0	29.5	29.9	30.4	29.3	26.5	25.3
		MCWBR	15.1	15.0	14.9	13.3	12.2	13.0	13.1	13.3	13.7	13.0	13.6	15.4
	5% WB	MCDBR	22.3	23.8	25.5	25.5	27.0	28.2	28.5	29.4	28.3	26.1	21.6	21.3
		MCWBR	14.5	14.5	14.3	12.9	12.4	12.9	13.4	13.7	13.7	12.5	12.0	14.7
Clear-Sky Solar Irradiance	taub	0.369	0.401	0.457	0.544	0.559	0.489	0.467	0.484	0.464	0.495	0.425	0.383	
	taud	2.092	1.993	1.840	1.671	1.657	1.811	1.887	1.848	1.876	1.812	1.953	2.051	
	Ebn at Noon	265	267	260	243	240	257	262	255	255	233	241	251	
	Edh at Noon	42	51	64	78	80	68	63	65	61	60	48	42	
All-Sky Solar Radiation	RadAvg	1144	1459	1837	2107	2347	2606	2502	2302	1990	1554	1209	1056	
	RadStd	52	81	57	72	93	54	75	55	68	69	96	46	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (1 neighbor)	+2.50	N/A	N/A	+1.98	N/A	N/A	-58	-257	+832	+551	

Nomenclature: See separate page